5 A Letter to Professor Bernhard Riemann

I was invited to give a talk in Varese, on June 4th of 2025, in the Villa Toeplitz, where the Riemann International School of Mathematics is located. In order to prepare this talk, I did, two days before, a pilgrimage in Selasca, the place where Riemann died on July 20th of 1866.
Given the place where I had to give the talk, I took the following challenge:

What could I tell to Riemann that would surprise him
and that would give him confidence that his hypothesis is true?

So what I am going to write now is a letter to him. I will address to him as "Master". The requisite of course is that it is forbidden to use mathematical notions unknown to him or not easily understandable by him. In fact, I will restrict myself to those which he used in his work. After a preliminary remark, the really new stuff is in the second part of the letter.

TO PROFESSOR BERNHARD RIEMANN

The beautiful formula that you proved in your paper, namely the formula which allows one to compute the number of prime numbers less than x in terms of the zeros of the zeta function, has been unfortunately badly written in textbooks thus displaying a profound misunderstanding of the middle term of the formula. You let $\pi'(x)$ be the number of primes strictly less than x with $\frac{1}{2}$ added when x is prime and found for the counting function

$$f(x) := \sum \frac{1}{n}\pi'\left(x^{\frac{1}{n}}\right),$$

the following formula involving the integral logarithm function $\mathrm{Li}(x) = \int_0^x \frac{dt}{\log t}$,

$$f(x) = \mathrm{Li}(x) - \sum_{\alpha}\left(\mathrm{Li}\left(x^{\frac{1}{2}+\alpha i}\right) + \mathrm{Li}\left(x^{\frac{1}{2}-\alpha i}\right)\right) + \int_x^{\infty} \frac{1}{t^2-1}\frac{dt}{t\log t} - \log 2 \quad (11)$$

You were very careful at the end of your paper in dealing with multivalued functions, but the mathematical language has evolved and, nowadays, one only uses functions with a definite univalent meaning, while the use of multivalued functions is no longer commonly admitted.
In standard textbooks, for instance the classic book of Edwards[7] the middle term is written as a sum of the integral logarithm Li function evaluated on x^ρ where the ρ's are the non-trivial zeros of ζ.
Now this is obviously a nonsensical sum because x to the power ρ does not change if one replaces ρ by $\rho + 2\pi i n/\log x$, where n is an arbitrary integer. So the constancy of $\mathrm{Li}(x^\rho)$ on this arithmetic progression implies that there is no possible meaning to the infinite sum because it includes infinitely many repetitions of essentially the same term, since the complex numbers x^ρ accumulate infinitely at least on the circle of radius $x^{1/2}$. In fact, von Mangoldt gave, many years after your paper, a detailed proof of your theorem, and he was more careful, up to some extent, in writing the formula ... since instead of dealing with Li he considers a well defined univalent function $\mathrm{Ei}(z) = \mathrm{Li}(\exp(z))$ of the correct variable z (rather than $\exp(z)$). He then proves your formula replacing the term $\sum \mathrm{Li}(x^\rho)$ by $\sum \mathrm{Ei}(\rho \log x)$.

This being said, it is clear from your formula that you knew what is now called the Guinand-Weil explicit formulas. And the new input in this formula, but that wouldn't surprise you, is that one considers, instead of the function $1/\log u$ which you use, an arbitrary test function $\phi(u)$ applied on the prime powers, and one writes an equality which involves what is nowadays called the Mellin transform[9] of this function $\phi(u)$ evaluated on the non-trivial zeros of the zeta function. One obtains in this way an equality which is not really a generalization of your formula in the sense that yours goes one level of subtlety beyond these classical Guinand-Weil explicit formulas since your test function $1/\log u$ is singular at $u = 1$. One thing which is striking in these explicit formulas is that if the test function $\phi(u)$ vanishes outside a finite interval $[1, x]$, then only finitely many primes are involved in its computation, those less than x. But this is of course the case for your formula which involves the function equal to $\phi(u) = 1/\log u$ in the interval $[1, x]$ and vanishing outside, so this would not be a surprise to you.

But let me tell you, Master, a fact which I find quite supportive of your hypothesis. The fact is the following. I will devise a process with which, by using only a few primes, I will be able to reach the first non-trivial zeros of your zeta function with remarkable precision. Now what do I mean by involving a few primes? I will take a concrete situation and I will imagine that I do not know what are the primes beyond 13. So I only know 2, 3, 5, 7, 11 and 13. Now at first it seems that in order to get any grasp on the zeros of zeta, it is necessary to involve all primes, because all of them are involved in the Euler product which is defining this function. And so it seems quite unbelievable that by truncating this Euler product, namely by only considering the terms in the Euler product which are involving the primes smaller than 13, one could get any hint on the zeros of the zeta function.

So let me describe the process using only notions that are familiar to you. Out of these primes, 2, 3, etc., up to 13, one fabricates a quadratic form. This quadratic form is similar to the quadratic form you were using when you applied the Dirichlet principle in proving the conformal mapping theorem. It is a quadratic form on the infinite dimensional space of functions $\phi(u)$ of a positive real variable which vanish outside the interval $[1, 13]$. The value $Q(\phi)$ of the quadratic form is obtained by applying the explicit formula to the function $\psi(v) = \int \phi(u)\phi(uv)du/u$. Thus because the function ϕ vanish outside the interval $[1, 13]$, the function ψ vanishes outside the interval $[1/13, 13]$ and one does not need to use any other prime power than $2, 3, 4, 5, 7, 8, 9, 11, 13$ to compute $Q(\phi)$. Next, I know how to prove that there is a function $\eta(u)$ realizing the minimum of the quadratic form $Q(\phi)$ while $\int \phi(u)^2 du/u = 1$. The proof of existence is entirely similar to the proof given by Hilbert in 1900, in his paper "Über das Dirichletsche Prinzip" for the Dirichlet principle. I then take the Mellin transform of the function $\eta(u)$. I also know how to prove[10] that the zeros of this Mellin transform are on the critical line[11]. This is proved modulo a condition of uniqueness of the minimum[12].

[9] The Mellin transform of $\phi(u)$ is defined as $\mathcal{M}(\phi)(s) = \int_0^\infty u^{s-1}\phi(u)du$

[10] See §6.1

[11] Which is normalized here as the imaginary line

[12] The proof uses a generalization of a Theorem of Caratheodory-Fejer on Toeplitz matrices, obtained in 1911, one needs to assume that the lowest eigenvalue of the quadratic form is simple and even

The amazing fact, which I want to report to you, is that when one computes—with the aid of modern computational machines far beyond what was available in your time—the zeros of the Mellin transform of $\eta(u)$, say its first 50 zeros, one finds an incredible coincidence with the non-trivial zeros of zeta. For instance, for the first zero, you find out that there are 54 decimal places which agree, and so on and so forth. The number of decimal places which agree decreases slowly, and when you reach the 50-th zero, only few decimal places agree.
I have computed these differences (upper bound of) between the values computed using primes less than 13 and the true values of zeros of zeta which I present herewith:

DIFFERENCES BETWEEN VALUES (USING PRIMES ≤ 13)

2.60179×10^{-55}, 4.80071×10^{-52}, 4.43756×10^{-50}, 3.89903×10^{-47}, 7.59453×10^{-46}, 1.13198×10^{-43}, 1.07245×10^{-41}, 1.2694×10^{-40}, 4.40141×10^{-38}, 4.24869×10^{-37}, 5.86724×10^{-36}, 3.24443×10^{-34}, 2.44517×10^{-32}, 9.02026×10^{-32}, 5.13539×10^{-30}, 7.04142×10^{-29}, 6.47754×10^{-28}, 4.96772×10^{-27}, 5.86016×10^{-25}, 3.76751×10^{-24}, 1.03779×10^{-23}, 3.52722×10^{-22}, 3.03977×10^{-21}, 5.66201×10^{-20}, 1.41755×10^{-19}, 2.19821×10^{-18}, 6.31599×10^{-17}, 1.42037×10^{-16}, 4.34328×10^{-16}, 4.47113×10^{-15}, 7.01522×10^{-14}, 3.81989×10^{-13}, 5.99581×10^{-13}, 4.26414×10^{-11}, 1.10653×10^{-10}, 1.95651×10^{-10}, 5.20728×10^{-10}, 2.05031×10^{-9}, 3.42274×10^{-8}, 2.10931×10^{-7}, 2.23714×10^{-7}, 5.95608×10^{-7}, 5.77737×10^{-6}, 0.000141389, 0.000556111, 0.000720794, 0.000314865, 0.0209081, 0.00313565, 0.00212727

What this is saying is that we have a firm grasp on your zeros, without at any point involving the infinity of the collection of all primes. And moreover we know a priori that all zeros of the Mellin transform of $\eta(u)$ are on the critical line.

What we do not know is that, when we increase the upper limit, which was $x = 13$ here, the corresponding set of zeros will converge towards the zeros of zeta. This is something which at this point is not proved. On the other hand, it seems that the abstract reason why your conjecture is true is now within reach since we know that the zeros of the Mellin transforms of the minimal eigenvectors η_x are purely imaginary, and we expect that the η_x converge to the function whose Mellin transform is your function $\Xi(it)$. The result would then follow from Hurwitz's theorem, which implies that all the zeros of the limit of a convergent sequence[13] of holomorphic functions whose zeros are on a fixed line are still on that line.

With respect and admiration,

Alain Connes

[13] uniformly on compact subsets